1. Research output list

I have authored 37 publications to date, including 28 peer-reviewed international journal articles (1 of which is a co-authored review) and 3 conference proceedings. Altogether these works have accrued over 1775 citations (inspirehep.net, as of October 16, 2026), with a corresponding h-index of 20. Unless noted otherwise, the research list below is ordered alphabetically. For each item, I contributed at first- or second-author level: I carried out the computations, co-led the writing, and developed the main ideas. The complete publication list can also be found at Inspire HEP inspirehep.net/authors/1639147.

1.1. Journal articles

[40] C. Benso, D. Bödeker, K. Kamada, K. Mukaida, L. Sagunski, **P. Schicho**, and K. Schmitz, *Baryon number freeze-out in the Standard Model, precisely*, (2026), [2609.07467].

[39] J. Chakrabortty, B. Siqueira Eduardo, S. Karmakar, **P. Schicho**, *Finite-temperature operator basis on $\mathbb{R}^3 \times S^1$ for SMEFT*, JHEP **09**, 241 (2026), [2605.02878].

[38] S. Biondini, **P. Schicho**, *A critical look at low-scale cosmological phase transitions in the PTA era*, (2026), [2607.02505].

[37] M. Kierkla, M. Lewicki, **P. Schicho**, D. Schmitt, and B. Swiezewska, *Theoretical uncertainties in reconstructing model parameters with gravitational waves from supercooled phase transitions*, (2026), [2607.18233].

[36] S. Bandyopadhyay, J. Chakrabortty, D. Dey, **P. Schicho**, and Tushar, *Higher-dimensional operators and Polyakov loop in hot Scalar QED from the heat kernel*, (2026), [2606.09779].

[35] F. Bernardo, R. G. Reinle, **P. Schicho**, *Matching higher-dimensional operators at finite temperature for general models*, (2026), [2605.15176].

[34] F. Bernardo, M. Chala, L. Gil, **P. Schicho**, *Hard thermal contributions to phase transition observables at NNLO*, JHEP **06**, 281 (2026), [2602.06962].

[33] A. Bhatnagar, D. Croon, **P. Schicho**, *Interpreting the 95 GeV resonance in the Two Higgs Doublet Model. Implications for the electroweak phase transition*, JHEP **03**, 014 (2026), [2506.20716].

[32] M. Kierkla, N. Ramberg, **P. Schicho**, and D. Schmitt, *Thermodynamical uncertainties for primordial black holes from cosmological phase transitions*, Phys. Rev. D **113**, 095024 (2026), [2506.15496].

[31] J. van de Vis, **P. Schicho**, L. Niemi, B. Laurent, J. Hirvonen, and O. Gould, *WallGo investigates: Theoretical uncertainties in the bubble wall velocity*, JHEP **04**, 041 (2026), [2510.27691].

[30] F. Bernardo, P. Klose, **P. Schicho**, and T. V. I. Tenkanen, *Higher-dimensional operators at finite temperature affect gravitational-wave predictions*, JHEP **08**, 109 (2025), [2503.18904].

[29] A. Ekstedt, O. Gould, J. Hirvonen, B. Laurent, L. Niemi, **P. Schicho**, and J. van de Vis, *How fast does the WallGo? A package for computing wall velocities in first-order phase transitions*, JHEP **04**, 101 (2025), [2411.04970].

[28] M. Kierkla, **P. Schicho**, B. Swiezewska, T. V. I. Tenkanen, and J. van de Vis, *Finite-temperature bubble nucleation with shifting scale hierarchies*, JHEP **07**, 153 (2025), [2503.13597].

[27] A. Ekstedt, **P. Schicho**, and T. V. I. Tenkanen, *Cosmological phase transitions at three loops: The final verdict on perturbation theory*, Phys. Rev. D **110**, 096006 (2024), [2405.18349].

[26] M. Lewicki, M. Merchand, L. Sagunski, **P. Schicho**, and D. Schmitt, *Impact of theoretical uncertainties on model parameter reconstruction from GW signals sourced by cosmological phase transitions*, Phys. Rev. D **110**, 023538 (2024), [2403.03769].

[25] J. Ghiglieri, **P. Schicho**, N. Schlusser, and E. Weitz, *The force-force correlator at the hard thermal scale of hot QCD*, JHEP **03**, 111 (2024), [2312.11731].

[24] C. Lwowski, K. Kaiser, J. Bucur, **P. Schicho**, and T. Kohnen, *Accuracy of using the axial length of the fellow eye for IOL calculation in retinal detachment eyes undergoing silicone oil removal*, Br. J. Ophthalmology (2023) 10.1136/bjo-2023-323581.

[23] J. Österman, **P. Schicho**, and A. Vuorinen, *Integrating by parts at finite density*, JHEP **08**, 212 (2023), [2304.05427].

[22] L. Sagunski, **P. Schicho**, and D. Schmitt, *Supercool exit: Gravitational waves from QCD-triggered conformal symmetry breaking*, Phys. Rev. D **107**, 123512 (2023), [2303.02450].

[21] G. Aarts, J. Aichelin, C. Allton, A. Athenodorou, D. Bachtis, C. Bonanno, N. Brambilla, E. Bratkovskaya, M. Bruno, M. Caselle, C. Conti, R. Contino, L. Cosmai, F. Cuteri, L. Del Debbio, M. D'Elia, P. Dimopoulos, F. Di Renzo, T. Galatyuk, J. N. Guenther, R. Houtz, F. Karsch, A. Y. Kotov, M. P. Lombardo, B. Lucini, L. Maio, M. Panero, J. M. Pawlowski, A. Pelissetto, O. Philipsen, A. Rago, C. Ratti, S. M. Ryan, F. Sannino, C. Sasaki, **P. Schicho**, C. Schmidt, S. Sharma, O. Soloveva, M. Sorba, and U.-J. Wiese, *Phase Transitions in Particle Physics: Results and Perspectives from Lattice Quantum Chromo-Dynamics*, Prog. Part. Nucl. Phys. **133**, 104070 (2023), [2301.04382].

[20] A. Ekstedt, **P. Schicho**, and T. V. I. Tenkanen, `DRalgo`*: A package for effective field theory approach for thermal phase transitions*, Comput. Phys. Commun. **288**, 108725 (2023), [2205.08815].

[19] T. Gorda, A. Kurkela, J. Österman, R. Paatelainen, S. Säppi, **P. Schicho**, K. Seppänen, and A. Vuorinen, *Degenerate fermionic matter at N3LO: Quantum electrodynamics*, Phys. Rev. D **107**, L031501 (2023), [2204.11893].

[18] T. Gorda, A. Kurkela, J. Österman, R. Paatelainen, S. Säppi, **P. Schicho**, K. Seppänen, and A. Vuorinen, *Soft photon propagation in a hot and dense medium to next-to-leading order*, Phys. Rev. D **107**, 036012 (2023), [2204.11279].

[17] J. Löfgren, M. J. Ramsey-Musolf, **P. Schicho**, and T. V. I. Tenkanen, *Nucleation at Finite Temperature: A Gauge-Invariant Perturbative Framework*, Phys. Rev. Lett. **130**, 251801 (2023), [2112.05472].

[16] **P. Schicho**, T. V. I. Tenkanen, and G. White, *Combining thermal resummation and gauge invariance for electroweak phase transition*, JHEP **11**, 047 (2022), [2203.04284].

[15] S. Biondini, **P. Schicho**, and T. V. I. Tenkanen, *Strong electroweak phase transition in t-channel simplified dark matter models*, JCAP **10**, 044 (2022), [2207.12207].

[14] J. Hirvonen, J. Löfgren, M. J. Ramsey-Musolf, **P. Schicho**, and T. V. I. Tenkanen, *Computing the gauge-invariant bubble nucleation rate in finite temperature effective field theory*, JHEP **07**, 135 (2022), [2112.08912].

[13] J. Ghiglieri, G. D. Moore, **P. Schicho**, and N. Schlusser, *The force-force-correlator in hot QCD perturbatively and from the lattice*, JHEP **02**, 58 (2022), [2112.01407].

[12] L. Niemi, **P. Schicho**, and T. V. I. Tenkanen, *Singlet-assisted electroweak phase transition at two loops*, Phys. Rev. D **103**, 115035 (2021), [2103.07467].

[11] D. Croon, O. Gould, **P. Schicho**, T. V. I. Tenkanen, and G. White, *Theoretical uncertainties for cosmological first-order phase transitions*, JHEP **04**, 055 (2021), [2009.10080].

[10] **P. M. Schicho**, T. V. I. Tenkanen, and J. Österman, *Robust approach to thermal resummation: Standard Model meets a singlet*, JHEP **06**, 130 (2021), [2102.11145].

[9] M. Laine, **P. Schicho**, and Y. Schröder, *A QCD Debye mass in a broad temperature range*, Phys. Rev. D **101**, 023532 (2020), [1911.09123].

[8] M. Laine, **P. Schicho**, and Y. Schröder, *Soft thermal contributions to 3-loop gauge coupling*, JHEP **2018**, 37 (2018), [1803.08689].

1.2. Conference proceedings

[7] J. Ghiglieri, G. D. Moore, **P. Schicho**, N. Schlusser, and E. Weitz. *Hard parton dispersion in the quark-gluon plasma, non-perturbatively.* In *11th International Conference on Hard and Electromagnetic Probes of High-Energy Nuclear Collisions* (July 2023). [2307.09297].

[6] M. Fraser, D. Björkman, K. Cornelis, B. Goddard, V. Kain, **P. Schicho**, C. Theis, and H. Vincke. *Modelling the Radioactivity Induced by Slow-Extraction Losses in the CERN SPS*. In *Proc. of International Particle Accelerator Conference (IPAC'17)* (May 2017), 1897–1900.

[5] M. A. Fraser, R. G. Alia, B. Balhan, H. Bartosik, C. Bertone, D. Björkman, J. Borburgh, N. Conan, K. Cornelis, L. Gatignon, B. Goddard, Y. Kadi, V. Kain, A. Mereghetti, F. Roncarolo, **P. M. Schicho**, J. Spanggaard, O. Stein, L. Stoel, F. M. Velotti, and H. Vincke. *SPS Slow Extraction Losses and Activation: Challenges and Possibilities for Improvement*. In *Proc. of International Particle Accelerator Conference (IPAC'17)* (Copenhagen. 2017), 611–614.

1.3. Theses

[4] **P. M. Schicho**, *Multi-loop investigations of strong interactions at high temperatures*, PhD thesis (U. Bern, 2020).

[3] **P. M. Schicho**, *Inhomogeneous condensation in quark-based QCD effective models via wavelet pseudoparticles*, MA thesis (ETH Zürich, 2016).

[2] **P. Schicho**, *π- and ρ-Meson mass spectroscopy from Lattice QCD*, BA thesis (TU Graz, 2014).

[1] **P. Schicho**, *Increasing the sensitivity of a search for supersymmetry in the single lepton channel with the Stransverse Mass*, Project thesis (HEPHY Vienna, 2014).

2. Presentations and conference contributions

Active participation as a speaker in the most important international conferences and workshops in the fields of particle cosmology, thermal QCD, and phase transitions. Since 2017, I have delivered a total of 12 invited conference talks, 21 contributed conference talks, and 39 seminars.

2.1. Invited talks at conferences and workshops

[W12] *Fun with higher orders and operators*, invited **plenary** talk at *1st Granada workshop on phase transitions*, Granada, Spain, 15/10/2026.

[W11] *Flavor effects in baryon number freeze-out in the Standard Model, precisely*, invited **plenary** talk at *PPC 2026*, Sydney, Australia, 01/09/2026.

[W10] *Probing early universe phase transition microphysics with gravitational waves*, invited **plenary** talk at *ATHEXIS – Athens Symposium on Exploring the Universe*, Athens, Greece, 08/06/2026.

[W9] *From microphysics to Gravitational Waves for cosmological phase transitions*, invited **plenary** talk at *the 4th Topics of Particle, Astro and Cosmo Frontiers (TOPAC 2026)*, Chengdu, China, 21/04/2026.

[W8] *Construction and limitations of EFTs for hot QFT*, invited **plenary** talk at *EFT, everywhere, all at once*, CERN, Geneva, Switzerland, 20/03/2026.

[W7] *Cosmological phase transitions as sources of gravitational waves*, invited **plenary** talk at *Gravitational Waves and the Early Universe: Accelerated Expansion, Dynamical Inhomogeneity, and Beyond*, KMI, Nagoya University, Japan, 13/03/2026.

[W6] *Perturbative cosmological phase transitions in a broad temperature range*, invited **plenary** talk at *The 2025 Beijing Particle Physics and Cosmology Symposium (BPCS 2025)*, Beijing, China, 27/09/2025.

[W5] *Perturbative cosmological phase transitions in a broad temperature range*, invited **plenary** talk at *Numerical simulations of early Universe sources of gravitational waves*, NORDITA, Stockholm, Sweden, 04/08/2025.

[W4] *Limits and limitations of perturbative approaches to cosmological phase transitions*, invited talk at *GW day at CERN*, Geneva, Switzerland, 05/03/2025.

[W3] *Reliable theoretical predictions for GWs from first-order phase transitions*, invited **plenary** talk at *Fun with the phases of the Universe*, IFPU, Trieste, Italy, 08/10/2024.

[W2] *Reliable theoretical predictions for GWs from first-order phase transitions*, invited **plenary** talk at *Fundamental physics and gravitational wave detectors*, Pollica physics center, Italy, 20/09/2024.

[W1] *Strong cosmological phase transitions through the EFT lens*, invited **plenary** talk at *Quarkonia meet Dark Matter*, Institute for Advanced Study, Technical University of

Munich, Germany, 19/03/2024.

[W0] *EFT framework for (precision) cosmological phase transition thermodynamics*, invited **plenary** talk at *What the heck happens when the Universe boils?*, Kavli IPMU, Tokyo, Japan, 05/12/2022.

2.2. Contributed talks

[C21] *Flavor effects in baryon number freeze-out in the Standard Model, precisely*, contributed talk at SEWM 2026, Helsinki, Finland, 20/08/2026.

[C20] *Thermodynamic limitations for primordial black holes from cosmological phase transitions*, contributed talk at NEHOP 2026 – New Horizons on Primordial Black Hole Physics, CERN, Switzerland, 18/06/2026.

[C19] *Higher-dimensional operators at finite temperature affect gravitational-wave predictions*, contributed talk at Extended Scalar Sectors From All Angles 2026, CERN, Switzerland, 04/06/2026.

[C18] *Uncertainty budgets for bubble wall velocities in cosmological phase transitions*, contributed talk at LISA Cosmology Working Group Workshop 2026, ICC, University of Barcelona, Spain, 03/06/2026.

[C17] *Uncertainty budgets for bubble wall velocities in cosmological phase transitions*, contributed talk at PLANCK2026 and 6th EuCAPT Symposium, CERN, Switzerland, 19/05/2026.

[C16] *Connecting micro- and intermediate scales in perturbative cosmological phase transitions*, contributed talk at Swiss LISA theory group meeting, UZH Zürich, Switzerland, 16/09/2025.

[C15] *Higher-dimensional operators at finite temperature affect gravitational-wave predictions*, contributed talk at Swiss Cosmology Days 2025, ETH Zürich, Switzerland, 06/06/2025.

[C14] *Higher-dimensional operators at finite temperature affect gravitational-wave predictions*, contributed talk at HEFT 2025, Geneva, Switzerland, 05/06/2025.

[C13] *Completing the perturbative program for cosmological phase transitions*, contributed talk at KEK theory meeting on particle physics phenomenology (KEK-PH2025winter), Tsukuba, Japan, 21/02/2025.

[C12] *Finite-temperature effective theories for first-order cosmological phase transitions*, contributed talk at High-Energy Theoretical Particle Physics Mini-Conference, Genève, Switzerland, 04/09/2024.

[C11] *Phase transition thermodynamic parameters at high precision*, contributed talk at 26th International Conference on Particle Physics and Cosmology (COSMO'23), Madrid, Spain, 14/09/2023.

[C10] *Degenerate fermionic matter at N^3LO*, contributed talk at ELEMENTS Annual Con-

ference, Bad Nauheim, Germany, 11/07/2023.

[C9] *Can EFT reveal if there was an electroweak phase transition?*, contributed talk at Cosmology from Home, 03/07/2023.

[C8] PRECLISA. *Gravitational waves from cosmological phase transitions: Precision cosmology in the light of LISA*, interview talk for the Ambizione grant of the Swiss National Science Foundation, Bern, Switzerland, 22/06/2023.

[C7] *Phase transition thermodynamic parameters at high precision*, contributed talk at How fast does the bubble grow?, DESY, Hamburg, Germany, 17/05/2023.

[C6] *Strong electroweak phase transition and simplified dark matter models*, contributed talk at Progress on Old and New Themes in cosmology (PONT) 2023, Avignon, France, 05/05/2023.

[C5] *Jet dispersion in hot QCD from the lattice*, contributed talk at SEWM 2022, Paris, France, 20/06/2022.

[C4] *(Non-)perturbative jet dispersion in hot QCD*, contributed talk at Quark Matter 2022, Kraków, Poland, 06/04/2022.

[C3] *(Non-)perturbative jet dispersion in hot QCD*, contributed talk at Mini workshop: Phase transitions in particle physics, Galileo Galilei Institute, Firenze, Italy, 30/03/2022.

[C2] *Soft thermal contributions to 3-loop gauge coupling*, contributed parallel talk at FunQCD (online), Barcelona, Spain, 29/03/2021.

[C1] *Fun with thermal dimension-six operators*, contributed parallel talk at SEWM 2018, Barcelona, Spain, 28/06/2018.

2.3. Seminar talks

[S39] *Baryon number freeze-out in the Standard Model, precisely*, invited seminar talk at Monash University, Australia, 23/09/2026.

[S38] *Finite-temperature bubble nucleation with shifting scale hierarchies*, invited seminar talk at University Melbourne, Australia, 21/09/2026.

[S37] *Finite-temperature bubble nucleation with shifting scale hierarchies*, invited seminar talk at University of Bologna, Italy, 14/05/2026.

[S36] *Finite-temperature bubble nucleation with shifting scale hierarchies*, invited seminar talk at Tor Vergata, Rome, Italy, 11/05/2026.

[S35] *Potentials, rates, and bubble velocities in cosmological phase transitions*, invited seminar talk at SISSA, Trieste, Italy, 04/05/2026.

[S34] *Cosmological phase transitions across all length scales*, invited seminar talk at UESTC, Chengdu, China, 22/04/2026.

[S33] *Microscopic descriptions of cosmological first-order phase transitions*, invited seminar talk at University of Stavanger, Norway, 09/04/2026.

[S32] *From microphysics to Gravitational Waves for cosmological phase transitions*, invited seminar talk at University of Oslo, Norway, 07/04/2026.

[S31] *Flavor effects on the baryon number freeze-out in the Standard Model, precisely*, invited seminar talk at Kobe University, Japan, 06/03/2026.

[S30] *Baryon number freeze-out in the Standard Model, precisely*, invited seminar talk at CERN, Geneva, Switzerland, 18/02/2026.

[S29] *Uncertainty budgets for bubble wall velocities in cosmological phase transitions*, invited seminar talk at Marietta Blau Institute for Particle Physics ÖAW, Vienna, Austria, 20/11/2025.

[S28] *Uncertainty budgets for bubble wall velocities in cosmological phase transitions*, invited seminar talk at ITP CAS, Beijing, China, 23/09/2025.

[S27] *Limits and limitations of perturbative approaches to cosmological phase transitions*, invited seminar talk at Warsaw University, Poland, 27/03/2025.

[S26] *Perturbative cosmological phase transitions and their bubble wall velocities*, invited seminar talk at Kavli IPMU, Tokyo, Japan, 26/02/2025.

[S25] *Completing the perturbative program for cosmological phase transitions*, invited seminar talk at Bielefeld University, Germany, 10/06/2024.

[S24] *Impact of computational diligence on GW signals from phase transitions*, invited seminar talk at Warsaw University, Poland, 23/11/2023.

[S23] *Impact of computational diligence on GW signals from phase transitions*, invited seminar talk at University of Basel, Basel, Switzerland, 12/10/2023.

[S22] *Impact of computational diligence on GW signals from phase transitions*, invited seminar talk at Helsinki Institute of Physics, Helsinki, Finland, 19/09/2023.

[S21] *Electroweak phase diagram at finite lepton number density*, contributed talk at Thermal Field Theory meets Phenomenology, Uppsala, Sweden, 15/06/2023.

[S20] *High-temperature effective field theories: Precision phase transition thermodynamics*, seminar talk at HEP seminar, Université de Genève, Switzerland, 30/05/2023.

[S19] *Integrating by parts at finite density*, informal talk at the DMGW meeting, Goethe University, Frankfurt, Germany, 10/05/2023.

[S18] *Degenerate fermionic matter at N^3LO*, invited seminar talk at Gravitation and Cosmology seminar, Utrecht University, Netherlands, 25/01/2023.

[S17] *What can EFT tell us about the electroweak phase transition?*, seminar talk at CRC-TR211 meeting and Colloquium, Bielefeld University, Germany, 24/01/2023.

[S16] *EFT framework for cosmological phase transition thermodynamics*, seminar talk at the AstroCoffee, Goethe University, Frankfurt, Germany, 08/11/2022.

[S15] *Degenerate fermionic matter at N^3LO*, invited seminar talk (online) at S@INT seminar, INT, Seattle, USA, 13/10/2022.

[S14] *(Gauge independent) Bubble nucleation rate at finite temperature*, invited seminar talk at Jožef Stefan Institute, Ljubljana, Slovenia, 15/09/2022.

[S13] *Can EFT tell us if there was an electroweak phase transition?*, invited seminar talk at University of Graz, Graz, Austria, 24/08/2022.

[S12] *Soft light-cone observables from electrostatic QCD*, invited seminar talk (online) at the QCD theory seminar, 11/07/2022.

[S11] *Degenerate fermionic matter at N^3LO*, invited seminar talk at the Nuclear Physics Colloquium, Goethe University, Frankfurt, Germany, 07/07/2022.

[S10] *Can EFT tell us if there was an electroweak phase transition?*, invited seminar talk at SUBATECH, Nantes, France, 16/06/2022.

[S9] *Electroweak phase transition: Combining thermal resummation and gauge invariance*, invited seminar talk at NICPB, Tallinn University, Estonia, 31/05/2022.

[S8] *Combining thermal resummation and gauge invariance for electroweak phase transition*, invited seminar talk (online) at School of Physics and Astronomy, Monash University, Australia, 24/05/2022.

[S7] *Effective theory approach to cosmological phase transitions*, invited seminar talk at Instituto de Astrofísica de Canarias, La Laguna, Spain, 03/03/2022.

[S6] *Gauge independent bubble nucleation rate at finite temperature*, invited seminar talk at University of Basel, Basel, Switzerland, 28/10/2021.

[S5] *Cosmological phase transition: Robust thermal resummation*, invited seminar talk at University of Bern, Bern, Switzerland, 19/10/2021.

[S4] *Cosmological phase transition: Robust thermal resummation*, invited seminar talk (online) at KIAS, Seoul, South Korea, 13/05/2021.

[S3] *How to be precise at the electroweak scale at finite temperature*, invited seminar talk (online) at Kavli IPMU, Tokyo, Japan, 25/11/2020.

[S2] *3-Loop Gauge Coupling in Hot Yang-Mills*, invited seminar talk at Helsinki Institute of Physics, Helsinki, Finland, 13/08/2019.

[S1] *Fun with thermal dimension-six operators*, invited seminar talk at Universidad del Bío-Bío, Chillán, Chile, 28/08/2018.